

CIND820: Big Data Analytics Project

Milestone 1 — Project Design

Responsible Credit Scoring:

Predictive Risk Modeling & Ethical Fairness Auditing

Submitted by:

Carlos Elizondo

Student Number:

500890062

Supervisor:

TBA

Date of Submission:

May 18, 2026

Toronto Metropolitan University



**Toronto
Metropolitan
University**

Table of Contents:

1. Professional Context and Problem Definition.....	3
1.1 Job-Simulation Track.....	3
1.2 Key Beneficiary.....	3
1.3 Problem Definition.....	3
1.4 Real-World Significance.....	3
2. Research Questions and Impact Hypothesis.....	4
2.1 Research Questions.....	4
2.2 Justification.....	4
2.3 Impact Hypothesis.....	4
3. Dataset, Data Biography, and Ethics Review.....	5
3.1 Dataset Overview.....	5
3.2 Data Biography.....	5
3.3 Dataset Characteristics and Limitations.....	5
3.4 Privacy and Ethics Audit.....	6
4. Proposed Technical Pipeline.....	7
4.1 Environment and Tooling.....	7
4.2 Stage 1 — Data Ingestion and Relational Aggregation.....	7
4.3 Stage 2 — Data Cleaning and Feature Engineering.....	7
4.4 Stage 3 — Feature Selection.....	7
4.5 Stage 4 — Supervised Learning: Model Tournament.....	8
4.6 Stage 5 — Unsupervised Learning: Customer Segmentation.....	8
4.7 Stage 6 — Risk-Based Pricing Framework.....	8
5. Responsible Credit Scoring: The Dual-Model Audit.....	10
5.1 Model A — Unconstrained Baseline.....	10
5.2 Model B — Ethically Constrained Model.....	10
5.3 Fairness Metrics.....	10
5.4 SHAP-Based Transparency and Proxy Leakage Detection.....	10
6. GenAI Declaration.....	11
6.1 Permitted Uses.....	11
6.2 Prohibited Uses.....	11
6.3 Verification Plan.....	11
References.....	12

1. Professional Context and Problem Definition

1.1 Job-Simulation Track

For this project I will be taking on the role of a Risk Intelligence Analyst in the Retail Banking & Financial Services Industry.

1.2 Key Beneficiary

The Chief Risk Officer (CRO) and Head of Lending Strategy at a Canadian retail bank are the primary audience for this project's outputs. This is because they are in charge of designing the system that assigns the interest rates given and whether a mortgage request would be approved or denied. A secondary beneficiary is the underserved applicant population. This could potentially include recent immigrants, young professionals, and low-income households. These groups can be excluded from credit markets not because they are bad borrowers, but because legacy scoring systems lack the data to evaluate them. In the current industry there is a very high importance on the credit score to decide whether a loan is risky or not.

1.3 Problem Definition

Legacy credit scoring models depend almost entirely on established credit history, which is reflected by their credit score. This system systematically locks out newcomers to Canada, young professionals, and low-income households who do not have long credit histories yet. For a bank, this creates two concrete problems: revenue left on the table from creditworthy applicants who are incorrectly declined, and growing regulatory exposure as automated systems face scrutiny for producing discriminatory outcomes. I will address this by building a predictive model that processes alternative behavioral data from Home Credit's relational tables. This will include aggregating installment payment histories, credit bureau records, and asset flags to capture creditworthiness that static, single-table models overlook. Beyond prediction, I will conduct a Responsible Credit Scoring audit by deliberately building two versions of the model: an unconstrained baseline that maximizes mathematical performance, and an ethically constrained version that removes proxy variables and meets regulatory fairness thresholds. Comparing the two is the core of what makes this project analytically meaningful.

1.4 Real-World Significance

This project is directly grounded in Canada's current regulatory environment. OSFI Guideline E-23, Model Risk Management, is enforced across federally regulated financial institutions, and its guideline includes machine learning models and automated decision systems in its definition of a model. (Office of the Superintendent of Financial Institutions, 2025). Banks must validate AI systems for proxy discrimination, maintain full explainability of model outputs, and govern the entire model lifecycle from development through monitoring. Deploying a credit risk model that cannot be audited for bias is a violation that exposes institutions like RBC, TD, and Scotiabank to formal remediation requirements. This project simulates precisely the kind of fairness-accuracy audit that a Model Risk Management team would be required to run before any ML-assisted lending tool is used in their decision making.

2. Research Questions and Impact Hypothesis

2.1 Research Questions

I structured the three questions starting with the data, moving through the model, and ending with a strategy that a lending team can actually act on

RQ1 (Exploratory: Feature Significance):

When a formal credit score is unavailable, which combination of behavioral indicators like payment history, installment patterns, and credit bureau records in addition to traditional demographic features such as income type and occupation, best predicts loan default risk among thin-file applicants?

RQ2 (Predictive: Feature Selection and Model Optimization):

Does a consensus feature selection framework where I combine the importance rankings from Chi-Squared tests, Pearson Correlation, Multiple Linear Regression, Naive Bayes, Decision Tree, and Random Forest into a voting table that retains the top features agreed upon across methods improve the predictive performance of Decision Tree, Naive Bayes, and Random Forest models compared to their all-feature baseline, as measured by accuracy, precision, recall, F1-score, AUC-ROC, and confusion matrix outcomes?

RQ3 (Prescriptive: Responsible Credit Scoring):

When the optimized consensus model is constrained to exclude features that function as proxies for protected characteristics under the Canadian Human Rights Act including age, gender, marital status, and family status (Human Rights Code, 1990) how does its predictive performance compare to the unconstrained optimized model, and what does the fairness trade-off, measured by Demographic Parity and Equalized Odds, reveal about the cost of responsible lending?

2.2 Justification

I mapped out these questions to replicate how a real risk analytics workflow actually runs: feature extraction and exploration first, predictive modeling and optimization second, and after the real world restraints of ethical practices when building systems for business decisions. Each question builds directly on the previous one, and all three are feasible within the course timeline given the dataset structure and planned tools.

2.3 Impact Hypothesis

Based on the dataset characteristics and the planned modeling approach, I expect the following outcomes:

- RQ1: Behavioral aggregates derived from the secondary tables such as average payment delay, proportion of installments missed, and maximum days past due will emerge as stronger predictors of default than static demographic features like income type and occupation, especially for thin-file applicants where the EXT_SOURCE scores are absent. The most predictive feature combination will be a mix of behavioral payment patterns and asset flags rather than any single category of variable.
- RQ2: The consensus feature selection framework will improve AUC-ROC and F1-Score across all three models relative to their all-feature baselines, by removing noise variables that dilute signal and reducing the risk of overfitting on sparse or imputed columns.

- RQ3: The ethically constrained Model B will still be statistically significant and can be used by financial institutions as a way to still predict who will default without including the categories protected under the human rights act. The SHAP comparison between the two models will reveal that much of Model A's demographic signal is carried by behavioral proxies, meaning the constrained model loses less than its missing features would suggest, because genuine payment behavior and protected-group membership are not as interchangeable as an unconstrained algorithm treats them.

3. Dataset, Data Biography, and Ethics Review

3.1 Dataset Overview

Dataset Name: Home Credit Default Risk

Source: Kaggle (<https://www.kaggle.com/c/home-credit-default-risk>)

Original Data Owner: Home Credit Group, an international consumer finance company specializing in lending to unbanked individuals across Asia and Eastern Europe.

License: Publicly released for educational and research use as part of a Kaggle competition. The dataset terms permit non-commercial academic and analytical use with attribution.

3.2 Data Biography

The Home Credit Default Risk dataset was originally compiled by Home Credit Group between 2007 and 2018 to challenge data scientists to build better credit models for underserved borrowers. It was released on Kaggle in 2018 as a competition dataset which has since then been completed and remains publicly accessible for educational purposes. I downloaded the data directly from the Kaggle platform.

The dataset is relational and it is made up of seven interconnected tables:

Table	Rows (approx.)	Description
application_train.csv	307,511	Main application data: demographics, assets, financials, TARGET variable
bureau.csv	1,716,428	Credit history from other institutions
bureau_balance.csv	27,299,925	Monthly balances of bureau credits
previous_application.csv	1,670,214	Prior loan applications at Home Credit
installments_payments.csv	13,605,401	Payment history on prior loans
POS_CASH_balance.csv	10,001,358	Monthly POS and cash loan balances
credit_card_balance.csv	3,840,312	Monthly credit card snapshots

3.3 Dataset Characteristics and Limitations

The main table holds 122 variables across 307,511 loan applications. The TARGET variable (1 = default, 0 = repaid) is heavily skewed to repaid. 92% paid and 8% defaulted. This means a model's accuracy will not be the best way to measure its effectiveness as NB could hit 92% accuracy by predicting 'no default' for everyone, while being completely useless for the bank's actual purpose. In addition, secondary tables add granular time-series records pushing total volume past 50 million rows across all files.

There are several limitations I will need to manage explicitly:

- Class imbalance: The 92/8 split between non-defaults and defaults poses a modeling challenge that requires specialized handling (see Section 4).
- Anomalous values: The DAYS_EMPLOYED column contains over 55,000 placeholder entries of 365,243, indicating retired or unemployed applicants encoded inconsistently.
- Missing data: Several columns, including OWN_CAR_AGE and EXT_SOURCE_1, are missing between 50–70% of values, requiring imputation strategy decisions.

3.4 Privacy and Ethics Audit

The dataset doesn't include names, addresses, or government IDs. That said, my third research question will be looking at keeping an ethical approach by not including the demographics protected under the Ontario Human Rights Code.

- CODE_GENDER is a direct protected attribute under Canadian human rights law.
- DAYS_BIRTH is used to calculate applicant age, also a protected characteristic under the Canadian Human Rights Act.
- NAME_INCOME_TYPE and ORGANIZATION_TYPE may function as socioeconomic or ethnic proxies depending on their correlation with protected groups.
- REGION_RATING_CLIENT is a geographic rating that may correlate with race or ethnicity if regions map onto historically segregated communities.

I will exclude all direct protected attributes from the feature set before training. During feature selection, I will run a proxy correlation audit using a correlation matrix to flag any remaining variables that correlate with protected attributes at $r > 0.70$. Every pruning decision will be logged, and those findings feed directly into the dual-model comparison in Section 5 since they are what operationally separates Model A from Model B.

4. Proposed Technical Pipeline

4.1 Environment and Tooling

I will run the entire pipeline in Google Colab, using Google Drive for persistent storage so the SQLite database survives between sessions. SQLite will handle the joining and collapsing of tens of millions of rows from the secondary tables. Once the data is down to the 307,000 applications on the primary table then I will select the features, build the voting table and rebuild the ethical model along with graphs for model comparison.

4.2 Stage 1: Data Ingestion and Relational Aggregation

I will load the raw CSVs from Google Drive into a SQLite database and join the seven tables on the primary key SK_ID_CURR. The output is a set of static behavioral features per applicant: average payment delay in days, maximum days past due, total

number of prior loans, proportion of payments missed. These behavioral aggregates are the variables that RQ1 is looking for.

Feature Leakage Prevention: One real risk at this stage is pulling in bureau or previous application records that were created after the loan decision date which is information that a loan officer would not have had in front of them. I will filter every secondary table join to include only records predating each application's DAYS_DECISION timestamp, keeping the pipeline temporally honest.

4.3 Stage 2: Data Cleaning and Feature Engineering

This stage works through the known quality issues in the dataset. I will flag and impute the DAYS_EMPLOYED placeholder anomaly (365,243 encodes 'retired' or 'unemployed', not a millennium of employment); cross-reference OWN_CAR_AGE against FLAG_OWN_CAR to resolve ownership missingness logically; downcast float64 and int64 columns to their 32-bit equivalents to keep Colab's RAM manageable; and engineer composite features like Debt-to-Income Ratio and Payment-to-Annuity Ratio that carry direct interpretive weight for the bank. I also need to know which 'allowed' features are secretly correlated with protected attributes before any model training begins, because those findings determine which features are eligible candidates for the RQ2 consensus selection framework and which must be quarantined for the RQ3 fairness audit.

4.4 Stage 3: Consensus Feature Selection

I will generate feature importance rankings from six independent methods: Chi-Squared test, Pearson Correlation, Multiple Linear Regression, Naive Bayes, Decision Tree, and Random Forest and consolidate them into a voting table. Features that appear in the top rankings across a majority of methods are retained while those that fail to achieve consensus are dropped. The rationale is that a feature earning consistent support across methods with different statistical assumptions is more likely to reflect genuine signal than one that scores well on only a single criterion.

4.5 Stage 4: Supervised Learning: Model Tournament

I will train three classification algorithms using both the full feature set and the consensus-selected feature set, so that RQ2 can be answered through direct before-and-after comparison:

Naive Bayes: Probabilistic baseline; computationally efficient but assumes feature independence.

Decision Tree: Captures non-linear, if-then relationships; produces visually interpretable rules.

Random Forest: Ensemble of decision trees; reduces overfitting, provides stable feature importance rankings for cross-validation against the consensus selection results.

Validation and Class Imbalance Strategy: To address the severe 92/8 class split I will use stratified sampling to guarantee the target ratio is precisely maintained across all training and validation folds. Logistic Regression will serve as the baseline industry benchmark for credit scoring. To account for the asymmetric financial risk of a false negative in lending, cost-sensitive learning via `class_weight='balanced'` will be applied to all models to penalize missed defaults more heavily than missed approvals. Models will be evaluated using AUC-ROC, F1-Score, and Partial AUC (pAUC at FRP < 0.1) to

prioritize risk mitigation without sacrificing credit volumes, with Precision-Recall curves serving as a supplementary diagnostic.

4.6 Stage 5: Unsupervised Learning: Customer Segmentation

After model training, I will apply K-Means Clustering to the applicant pool to surface the distinct risk personas implied by RQ1's behavioral feature findings. The optimal number of clusters will be determined using the Elbow Method and Silhouette Score. I will then profile each cluster using mean feature values and assign descriptive labels. I expect to find at least three meaningful personas: a 'Struggling Starter' (young, short tenure, high education but no track record), an 'Over-Leveraged Professional' (moderate income, high debt-to-income ratio, multiple prior loans), and an 'Inconsistent Payer' (stable employment but a pattern of late installment payments). These personas ground the project's findings in human terms that a lending team can act on, and they inform the interpretation of which behavioral features drive default risk across meaningfully different borrower profiles.

4.7 Feasibility Note

The full pipeline runs within a standard Google Colab environment using open-source libraries (Scikit-Learn, Pandas, SHAP, Fairlearn or AI Fairness 360). I will validate each stage on a 10% sample before committing to a full-dataset run, which keeps iteration fast and avoids wasting compute hours on code that has not been verified.

5. Responsible Credit Scoring: The Dual-Model Audit

The defining analytical contribution of this project is the audit that RQ3 demands. I will build two versions of the same optimized pipeline and compare them directly, so the project produces a measured argument about the cost of responsible lending rather than simply a model.

5.1 Model A: Unconstrained Baseline

Model A trains on the full consensus-selected feature set. This model's only objective is performance. I expect it to achieve a high AUC-ROC, but I also expect its SHAP values to surface features that correlate with protected attributes. Model A documents what an unconstrained, well-optimized algorithm naturally learns to rely on when given no ethical guardrails, establishing the performance ceiling against which Model B is judged.

5.2 Model B: Ethically Constrained Model

Model B applies three interventions before training: removal of all protected attributes and proxy variables flagged in the Section 3.4 audit; threshold calibration to equalize approval rates across demographic segments; and a documented log of every pruning decision with its justification. This is the version that could realistically go in front of an OSFI Model Risk reviewer under Guideline E-23. The comparison between Model A and Model B is what answers RQ3: how much predictive accuracy, if any, does responsible modeling actually cost, and is that cost justifiable given the fairness gains measured by Demographic Parity and Equalized Odds?

5.3 Fairness Metrics

I will compute the following fairness metrics for both models using Fairlearn or AI Fairness 360:

- Demographic Parity: Measures whether approval rates are equal across groups.
- Equalized Odds: Measures whether True Positive and False Positive rates are equal across groups.

- Disparate Impact Ratio: Regulatory benchmark (minimum 0.80) for protected group approval rates relative to the majority group.

5.4 SHAP-Based Transparency and Proxy Leakage Detection

Model Interpretability and Proxy Audit (RQ3):

To explain how the models actually make decisions and answer RQ3, I will use SHAP values (Lundberg & Lee, 2017) to create two types of visualizations:

- Global Summary Plots: These show which features are most important across the entire population (e.g., income vs. credit history).
- Local Force Plots: These act like an audit trail for individual applicants, showing exactly why a specific person was approved or denied.

Proxy Leakage Check:

When comparing Model A and Model B side-by-side, I will check if the "ethical model" (Model B) is secretly using proxy variables to discriminate. For example, if after I remove gender, the model might try to use occupation or industry as a hidden substitute for gender. If SHAP values reveal that these placeholder features are acting as strong substitutes for hidden demographic traits, they will be flagged and removed. This ensures the model relies on true financial behavior rather than disguised demographic data.

6. GenAI Declaration

- No synthetic data will be generated or used. All training and evaluation will occur on the original Home Credit dataset as published on Kaggle.
- I will not use AI tools to generate final model outputs, evaluate results, or produce analytical conclusions.
- Any use of AI will be documented and any prompts used will be in the same document to make sure that I follow TMU's rules and policies.

References

- Bellamy, R. K. E., Dey, K., et al (2018). AI Fairness 360: An extensible toolkit for detecting, understanding, and mitigating unwanted algorithmic bias. arXiv preprint arXiv:1810.01943. <https://doi.org/10.48550/arXiv.1810.01943>
- Bird, S., Dudík, M., Edgar, R., Horn, B., Lutz, R., Milan, V., Sameki, M., Schuller, H., & Walker, K. (2020). Fairlearn: A toolkit for assessing and improving fairness in AI. Microsoft. <https://www.microsoft.com/en-us/research/publication/fairlearn-a-toolkit-for-assessing-and-improving-fairness-in-ai/>
- Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Dwork, C., Hardt, M., Pitassi, T., Reingold, O., & Zemel, R. (2012). Fairness through awareness. Proceedings of the 3rd Innovations in Theoretical Computer Science Conference, 214–226. <https://doi.org/10.1145/2090236.2090255>
- Human Rights Code, R.S.O. 1990, c. H.19. Ontario Legislature. <https://www.ontario.ca/laws/statute/90h19>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30. <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- McKinney, W. (2010). Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference, 56–61. <https://doi.org/10.25080/Majors-92bf1922-00a>
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. ACM Computing Surveys, 54(6), 1–35. <https://doi.org/10.1145/3457607>
- Montoya, A., Inversion, Odintsov, K., & Kotek, M. (2018). Home Credit Default Risk [Kaggle competition]. <https://kaggle.com/competitions/home-credit-default-risk>
- Office of the Superintendent of Financial Institutions. (2025, September 11). *Guideline E-23 – Model Risk Management (2027)*. Government of Canada. <https://www.osfi-bsif.gc.ca/en/guidance/guidance-library/guideline-e-23-model-risk-management-2027>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- SQLite Development Team. (2024). SQLite (Version 3.45) [Software]. <https://www.sqlite.org/>